

2024 Linguistic Summer Institute of Taiwan

Coordination

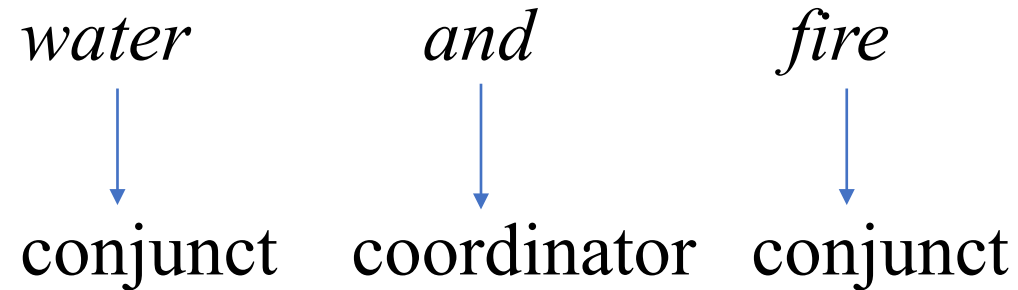
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What is coordination?

A coordinate construction, such as



contains at least two conjuncts, and an overt or silent coordinator, such as the conjunction *and*, the disjunctive *or*, or the adversive joining marker *but*.

What are we going to learn in this course?

How is a coordinate construction built in syntax?

What is the syntactic relation between conjuncts?

What are the relation between conjuncts and coordinators?

What are the properties of coordinators?

The syntax of coordination is naturally related to the syntax of modification.

Steps of this course

1. The basic link between a coordination and a modification construction
2. The systematic similarities of the two constructions
3. The systematic similarities of two kinds of functional elements
4. The CSC issue
5. Three types of the ATB constructions
6. Significance

1. The basic link between a coordination and a modification construction

Let us start with the modification part

Q1: What is the syntactic relation between a modifier and its modified element?

Q2: What are the properties of a MOD marker?

Let us start with Q1:

Generally speaking, a modifier is optional to its modified element; but if a modifier occurs, a modified element must occur.

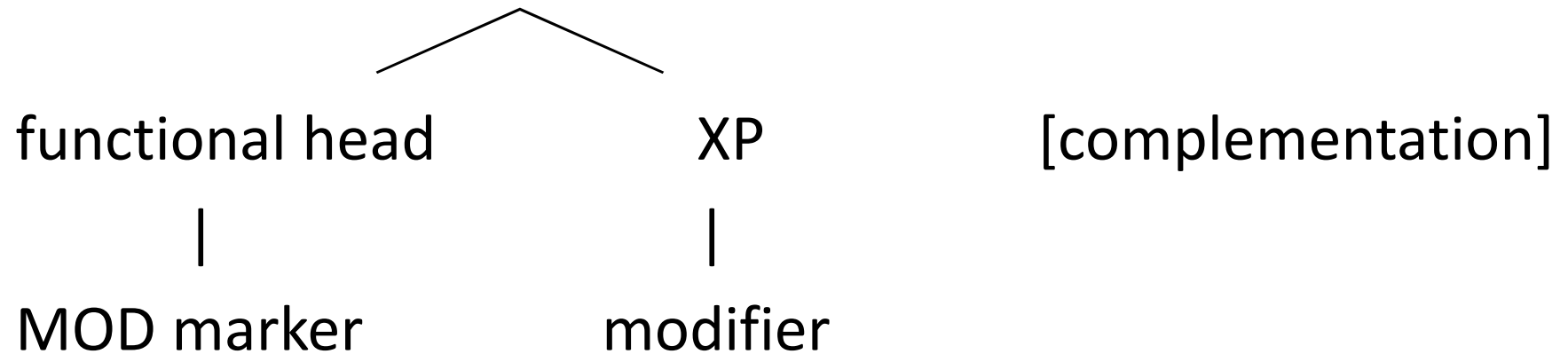
=> The occurrence of a modifier depends on the occurrence of a modified element, but not vice versa.

How do we represent this dependency in syntax?

- For a typical transitive verb, it requires the occurrence of an object.
- Usually such a verb is merged with its object directly.
- If a modifier needs the occurrence of a modified element, is the modifier merged with the modified element directly?
- In some languages, there is a MOD marker between a modifier and a modified element (胖胖的小孩 'fat MOD kid') (Rubin 2003).
- If the latter two are merged directly, where is the position of the MOD marker?
- Let us examine the properties of MOD markers (Q2).

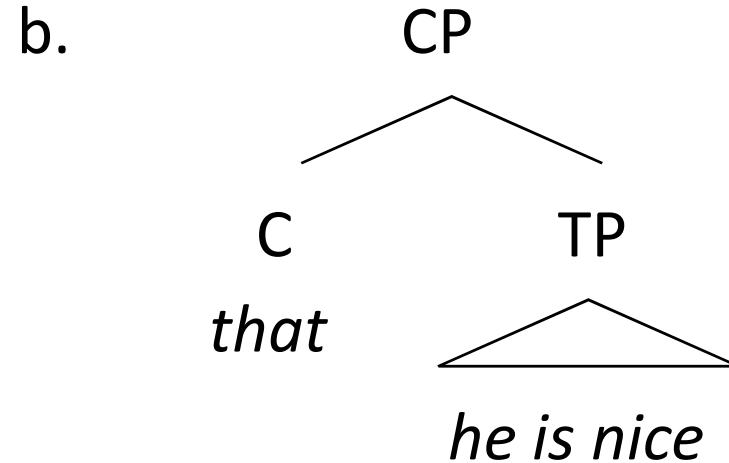
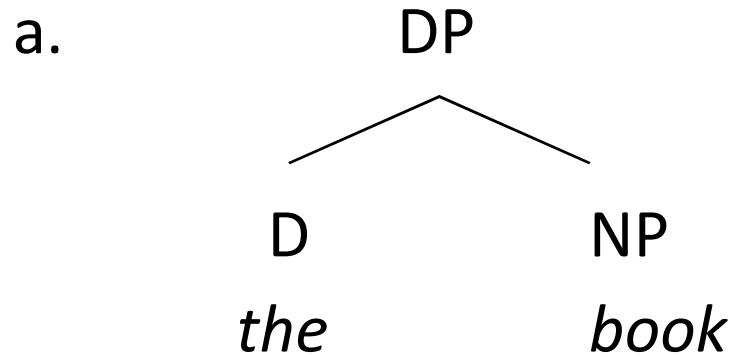
MOD markers

- Languages such as English do not have overt MOD markers.
- In Mandarin, the MOD marker *de* introduces a modifier to a modified element.
- A MOD marker requires the occurrence of a modifier.
- There is no semantic or categorial selection of a MOD marker on a modifier, unlike the selection relation between a transitive verb and its object.
- The relation between a MOD marker and a modifier is more like that between a functional head and its complement: the former must be merged with a complement.



- Then, where is the modified element, if it is not merged with the modifier directly?
- Why does a modifier need a modified element in syntax?
Is this because of the functional head?

But:



Why is the combination of a MOD marker and a modifier never complete?

Does a MOD marker have an EPP feature? Stipulating an obligatory Spec is not an account.

My answer: such a functional element is defective in a certain way:

it needs a support because it cannot project categorial features, unlike D and C.

A MOD marker plays no role in the category of either the modifier or the whole construction

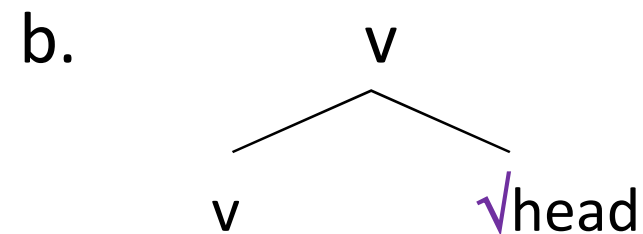
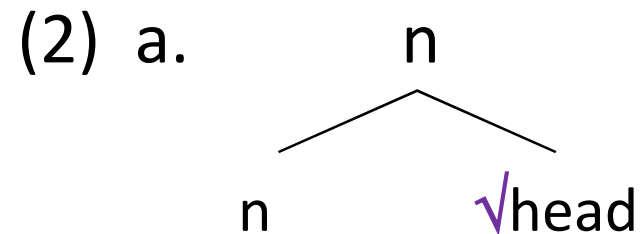
- (1) a. Yani mei kanjian qita-de shangdian. modifier XP
Adj NP
Yani not see other-MOD shop
'Yani did not see the other shops.'
- b. Na-ge likai-de nanhai you lai-le. *CP NP*
that-CL leave-MOD boy again come-PRF
'The boy who had left came again.'
- c. Yani kanjian-le ji-jia yan malu-de shangdian. *PP NP*
Yani see-PRF several-CL along road-MOD shop
'Yani saw several shops along the road.'
- (2) Yani qiaoqiao-de likai-le. *Adv VP*
Yani quietly-MOD leave-PRF
'Yani left quietly.'

Hypothesis: MOD markers have no categorial features

- They are thus similar to roots, in this respect.
- A root has no category.

(1) a. the **head** of this frog

b. She **headed** to the door.

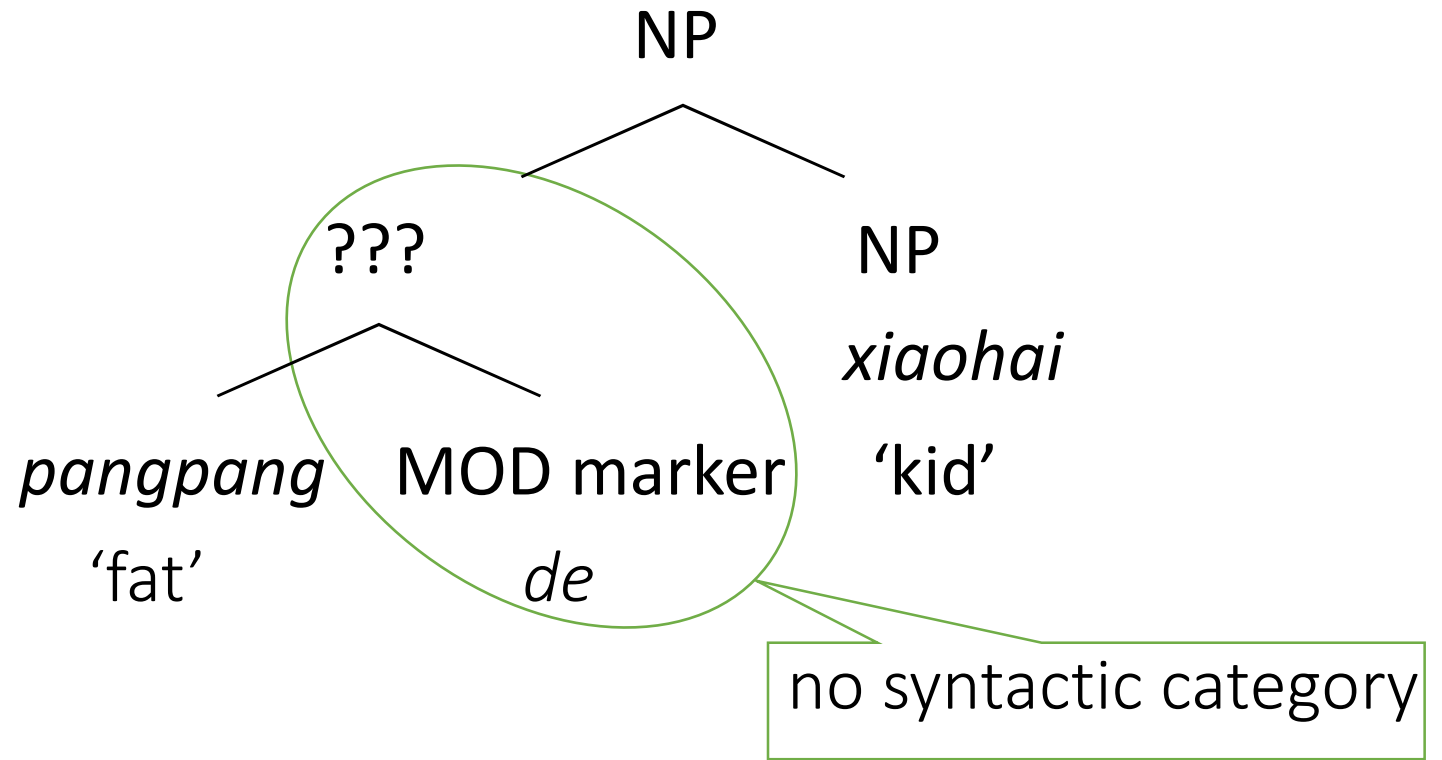


- MOD markers are thus similar to roots, in this respect.
- If so, the combination of a MOD marker and a modifier also has no categorial features, since a complement does not project.
- If so, some other element must categorize the combination.
- That is the job of the modified element.
- Two layers of external merge: complementation (low) and categorization (high).

Two layers of external merge in modification

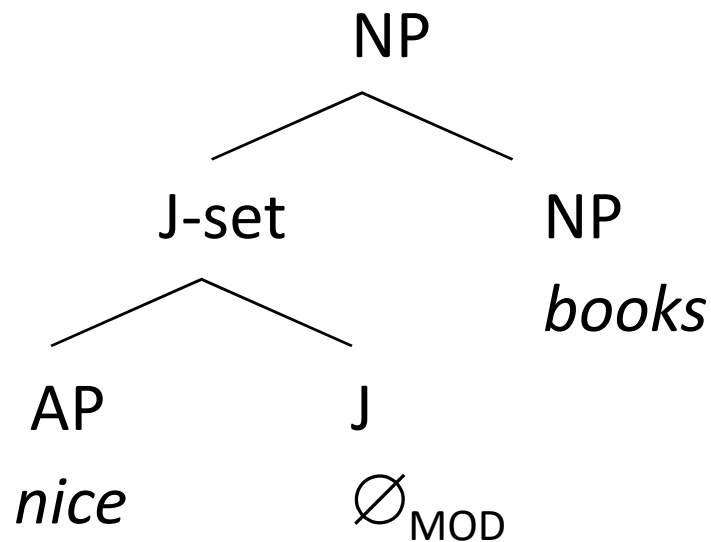
Categorization ➤

Complementation ➤



Two necessary levels of external merge in modification

- All modifiers are introduced by the functional element Junct (J).
- J has no category features.
- J can be realized by a MOD marker.



➤ The NP categorizes the J-set.

➤ J takes a complement

In each level of the two merges, the order is irrelevant (to be elaborated later).

Are you ready to answer some questions?

(I)

Q1: What is the shared property of J and a root?

Q2: What does the sister of a J-set do?

Q3: Is a modifier a sister of the modified element?

Are you ready to make some comments on the analysis so far?

We are now back to the coordination issue

Q1: What is the syntactic relation between conjuncts and a coordinator?

Q2: What are the properties of a coordinator?

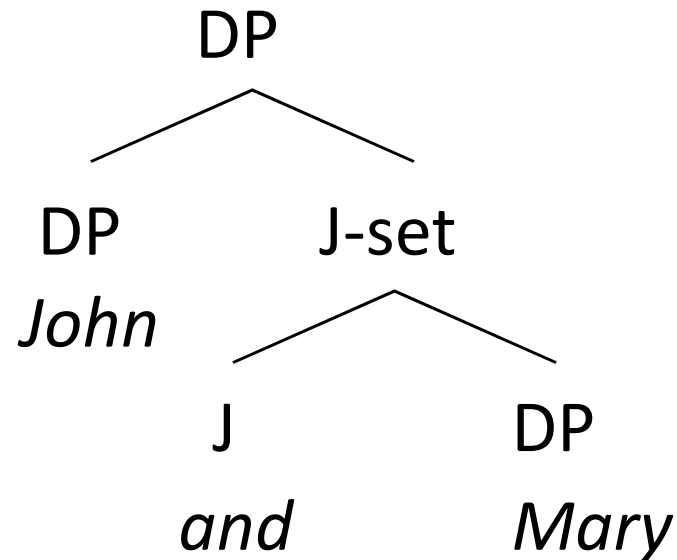
Let us start with Q1: when one conjunct is merged with a coordinator, their combination must be merged with another conjunct.

Let us call the former conjunct internal conjunct and the latter one external conjunct.

If merge is always binary (Collins 1997), the internal conjunct is lower than the external one.

Two necessary levels of external merge in coordination

- An internal conjunct, like a modifier, is introduced by J.
- J can also be realized by a coordinator (as well as a MOD marker).
- J has no category features (to be shown next).



➤ The DP categorizes the J-set.

➤ J takes a complement

How do we know that a coordinator has no category features?

Recall: a MOD marker has no category features.

The same *de* occurs in both a nominal and a verbal modification construction.

It is the modified element that decides the category of the whole modification construction.

Similarly, a coordinator has no category features.

The same *and* occurs in both a nominal and a verbal coordinate construction: *books and magazines; sing and dance; small and big*.

It is the external conjunct that decides the category of the whole coordinate construction.

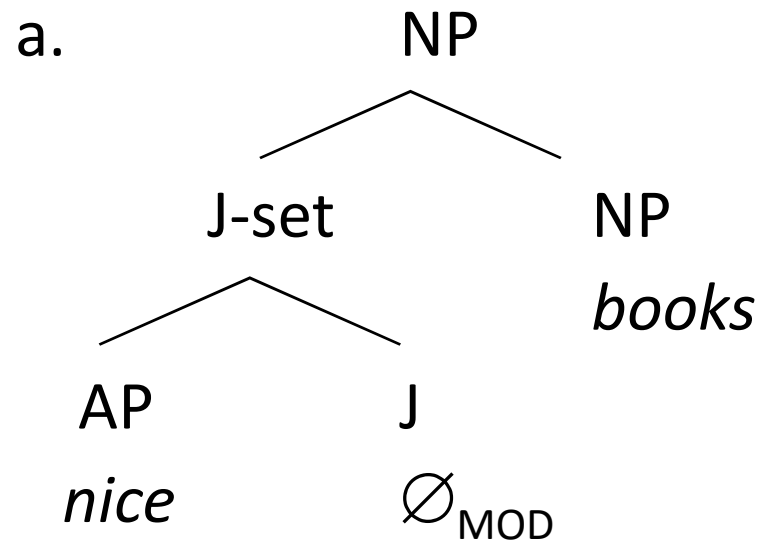
Answer to Q2: A coordinator realizes the categoryless functional element J.

But why are there contrasts between various coordinators in Mandarin?

- a. Alan xihuan he pijiu {gen/*you} lū-cha.
Alan like drink beer and/and green-tea
'Alan likes to drink beer and green-tea.'
- b. Alan shanliang {you/*gen} youmo.
Alan kind and/and humorous
'Alan is kind and humorous.'

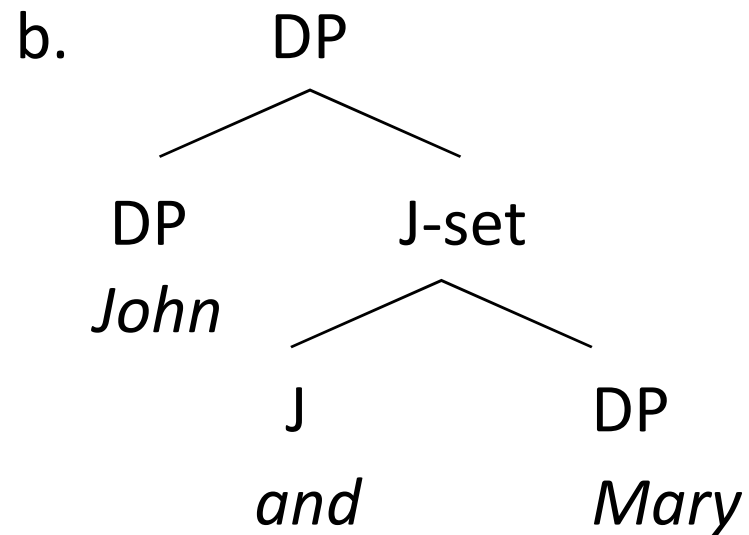
- Coordinators may have c-selection.
- The fact that X c-selects Y does not mean that X is also Y.
- This c-selection fact does not falsify that it is the external conjunct that decides the category of the whole coordinate complex.
- Whether an element has a category and whether it has c-selection are different issues.

2. The systematic similarities of the two constructions



➤ The NP categorizes the J-set.

➤ J takes a complement



➤ The DP categorizes the J-set.

➤ J takes a complement

- Five points

2.1 The categorial decisiveness of the higher item

- The modified element decides the category of the whole construction.
 - The external conjunct decides the category of the whole construction.
- (1) a. You can depend on [_{NP} my assistance] and [_{CP} that he will be on time].
b. *You can depend on [_{CP} that he will be on time].
c. *You can depend on [_{CP} that he will be on time] and [_{NP} my assistance].
 - (2) a. We talked about Mr. Golson's many qualifications and that he had worked at the White House.
b. *We talked about that he had worked at the White House.
 - (3) a. Pat was annoyed by [the children's noise and that their parents did nothing to stop it]
b. *Pat was annoyed by that their parents did nothing to stop it. (Sag *et al* 1985)
 - (4) a. John devoured [_{NP} only pork] and [_{PP} only at home].
b. ?*John devoured [_{PP} only at home] and [_{NP} only pork].

2.2 Syntactic optionality of the lower item

It is well-recognized that a modifier is optional in general.

- (1) a. I met a (tall) boy.
- b. I met a boy (and a girl).

2.3 Stackability

- (1) I met a young, happy, diligent, . . . , and smart boy.
- (2) a. John is tall, happy, hungry, bored with TV, etc.
 b. I met someone young, happy, eager to go to college, tired of wasting time, . . .

Chomsky (2013): unbounded unstructured coordination construction

2.4 Alternation Possibilities between Internal Conjuncts and Modifiers

A modifier may have a conjunctive reading.

Reading B of (1) is synonymous with the coordinate construction (2) (Progovac 1999).

(1) I read his paper quickly.

Reading A: 'I engaged in a quick reading of his paper.'

Reading B: 'There is an event of reading of the paper by me, and the event was quick.'

(2) I read his paper, and quickly.

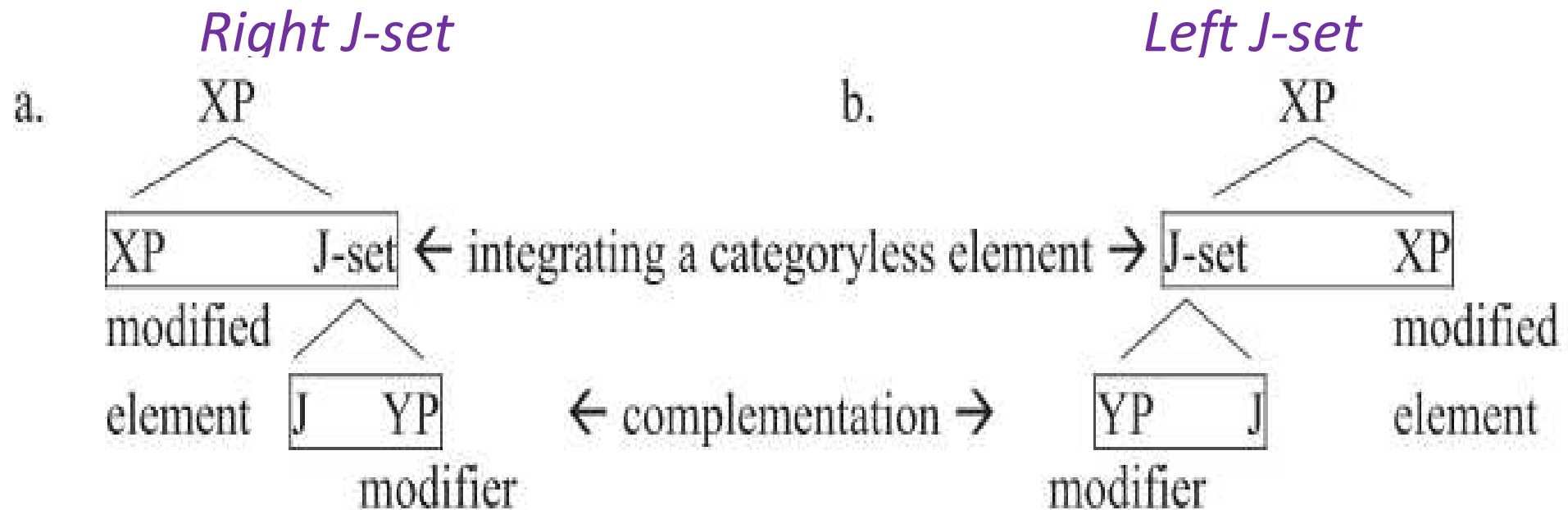
Moreover, an adverbial clause introduced by the temporal use of the complementizer *while* can get a conjunctive reading in examples like (1b) and (2b), similar to the coordinate constructions in (1a) and (2a), respectively (Haegeman 2022).

- (1) a. John is doing a Ph.D. in Oxford **but** Bill did his first degree in Cambridge.
b. John is doing a Ph.D. in Oxford **while** Bill did his first degree in Cambridge.

- (2) a. John reads the Guardian **and** Mary reads the Times.
b. John reads the Guardian **while** Mary reads the Times.

2.5 The Position Flexibility of the J-set

- Modification: (1) a. I opened the door **quickly**.
b. I **quickly** opened the door.



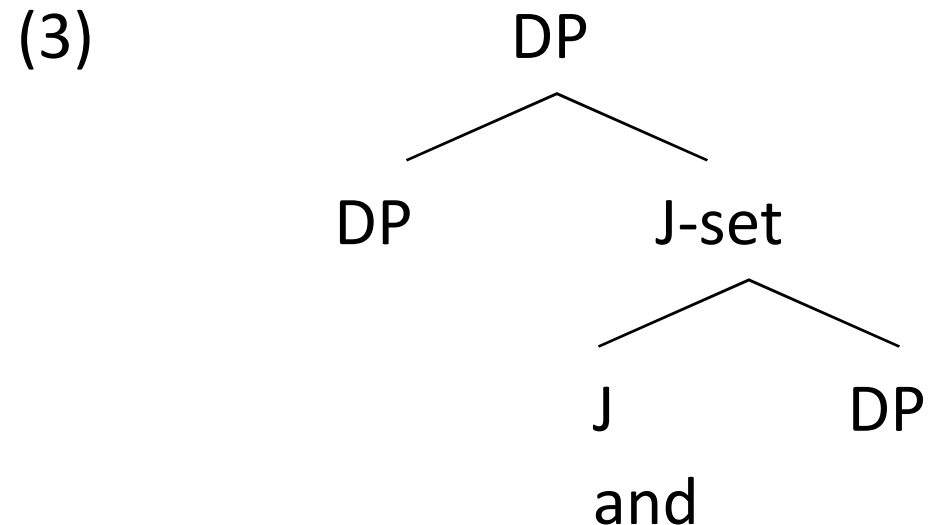
Coordination: also two possibilities

Nominal Coordination: *Right J-set only*

Two contrasts: pronominalization and binding

- (1) a. John_i's dog and he_i went for a walk.
b. *He_i and John_i's dog went for a walk. (Cond. C violation)

- (2) a. every dog_i and his_i owner
b. *his_i owner and every dog_i (Cond. A violation)



Verbal Coordination:

Pattern A: the left conjunct has a conditional reading: *Left J-set*

(2) You drink one more can of beer **and** I'm leaving.

(Culicover & Jackendoff 1997)

(= **If** you drink another can of beer, I'm leaving.)

=> The right conjunct c-commands the left one.

Pattern B: Elsewhere: *Right J-set* is possible

Also two contrasts:

Backward pronominalization from the right clause

A: The left-clause has a conditional reading:

- (1) a. [If you give it₁ fresh fish], [a cat₁ will love you forever]. *Adverbial*
b. [Give it₁ fresh fish], and [a cat₁ will love you forever]. *&*

B:

- (2) a. [Some woman₁ sat at a table] and [the host offered her₁ drinks].
b. *[She₁ sat at a table] and [the host offered some woman₁ drinks].

Binding

(cf. Privoznov 2021; Culicover & Jackendoff 1997)

A: The left-clause has a conditional reading:

(1) a. **If** you give **it**₁ fresh fish, [every cat₁, no matter how wild it is, will love you forever]. *Adverbial*

b. Give **it**₁ fresh fish, **and** [every cat₁, no matter how wild it is, will love you forever]. *&*

B:

(2) *You gave **it**_i fresh fish **and** every cat₁ has been locked up.

The contrasts between Pattern A and Pattern B

	Backward pronominal- ization	Binding from the right clause
Conditional Adverbial	√	√
Pattern A left-conjunct	√	√
Pattern B left conjunct	No	No

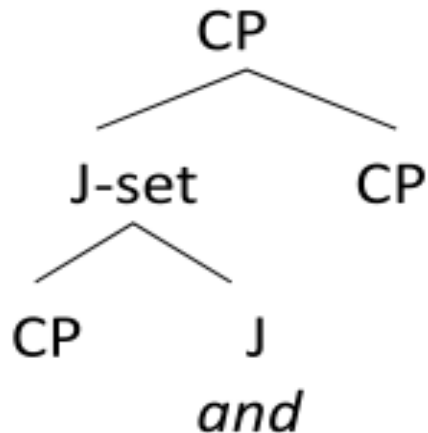
What do the contrasts tell us?

Not all coordinate constructions have the same structure.

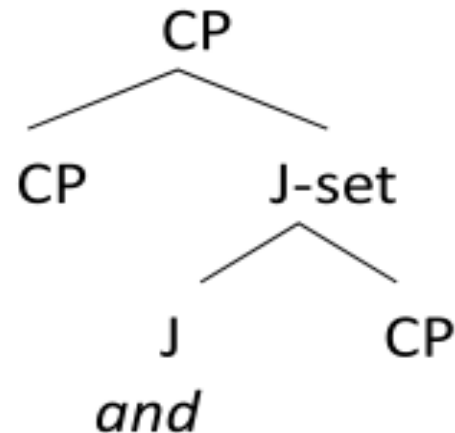
The left conjunct has a subordinate reading.

The two conjuncts are symmetrical or the right one has a subordinate reading

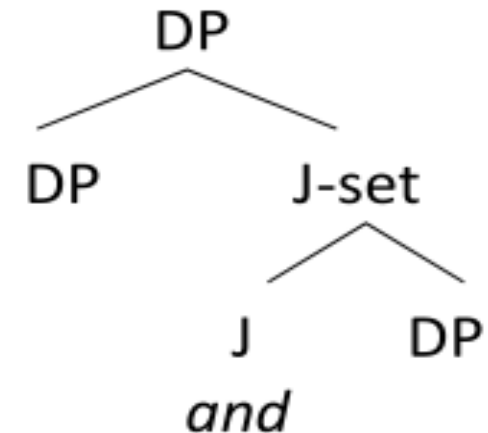
(1) a. Pattern A



b. Pattern B

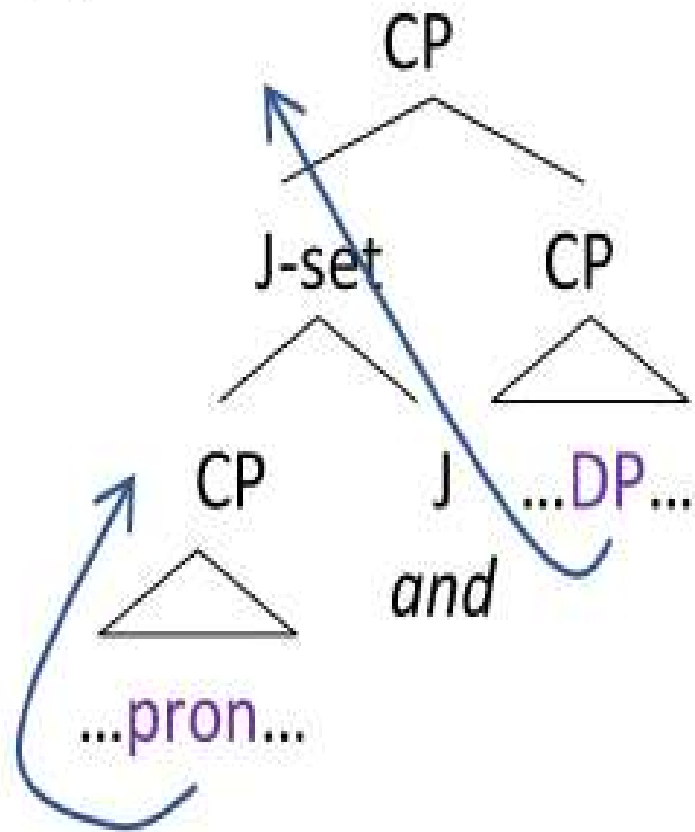


c. Nominal Coordination

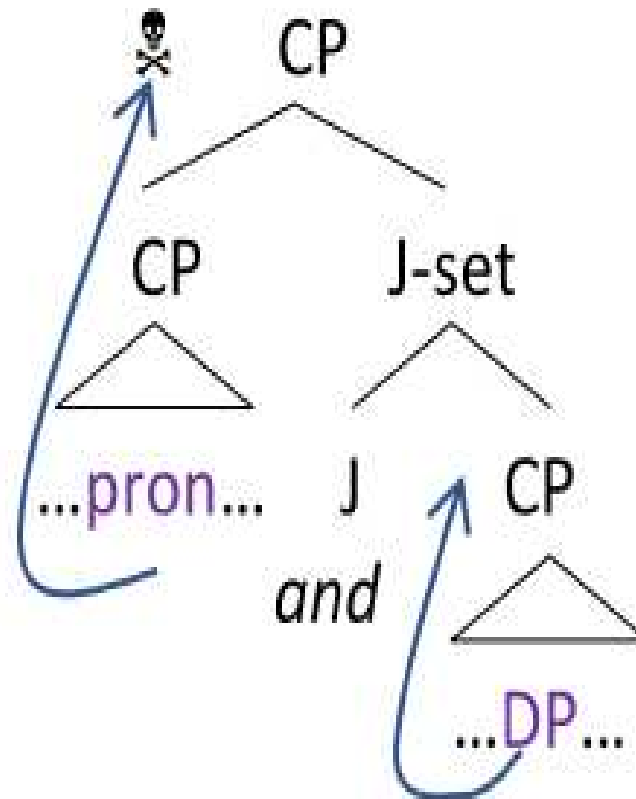


One possible account: c-commanding at LF, in the clausal coordination.

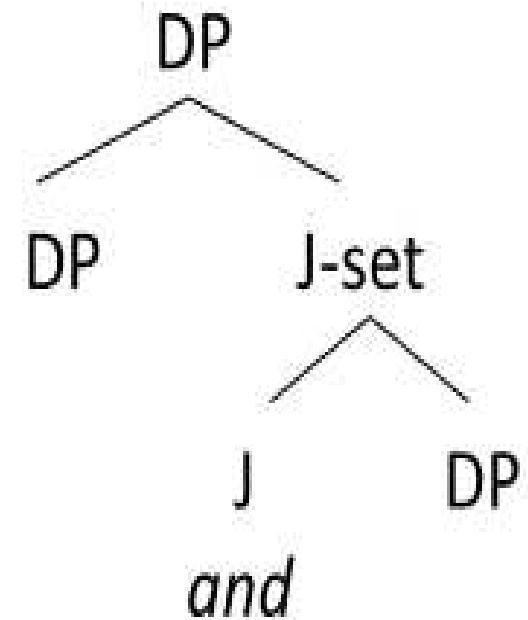
(1) a. Pattern A



b. Pattern B



c. Nominal Coordination



Are you ready to answer some questions?

(II)

Q1: We have introduced a few similarities between a coordinate construction and a modification construction. Tell us one of them.

Q2: Is a nominal coordination left-branching or right-branching?

Q3: Is a clausal coordination left-branching or right-branching?

Are you ready to make some comments on the analysis so far?

3. The systematic similarities of two kinds of functional elements (coordinators and MOD markers)

First, like some pure functional elements such as C and T, neither has a theta relation with another element.

Second, like other functional elements, both are a closed set.

Third, both must be associated with at least two items: one links a modifier to a modified element, and the other links one conjunct to another conjunct. => Two levels of external merge

Fourth, neither has any category features. *And* does not project any categorial features, nor does the MOD marker *de* in Mandarin (Li 2008).

Fifth, like many functional heads, both can be silent in a language or in certain constructions of a language.

J is null in MOD constructions in English.

In Mandarin, a null MOD marker is seen in examples like (1).

- (1) Congming ren bu gan na-zhong shi.
wise person not do that-kind thing
'Wise people do not do that kind of thing.'

Null coordinator constructions:

- a. I found no more than two, three mistakes in your article. (Neeleman et al. 2023: 74)
- b. They went, the one to Paris, the other to London. (Kayne 1994: 144)
- c. Baoyu Daiyu, ni xihuan na yi-ge? [Mandarin]
Baoyu Daiyu you like which one-CL
'Baoyu {and/or} Daiyu, which one do you like?'
- d. Qichuang-hou dajia dou hui shua ya xi lian.
get.up-after everyone all will brush tooth wash face
'After getting up, everyone will brush the teeth and wash the face.'
- e. Mei-jia zhi neng lai yi-ge ren, daren xiaohai dou xing.
each-family only can come one-CL person adult child all OK
'Only one person from each family is allowed to come. Either an adult or a child is OK.'

Sixth, both appear in a linker position.

- If two elements have a morpho-syntactic dependency, the dependency relation can be marked by an overt form.
- Such an overt form can be morphologically attached to the dependent element or the other element, called Head (Nichols 1986).

a. ^M**d-ovxa** ^Hxi [Chechen]
hot water(d-)
'hot water'

b. ^Hkũh-^Mi baland [Tadzhik]
mountain high
'high mountain'

- In addition to the above two patterns of dependency marking, a third pattern is called neutral marking.
- In this pattern, a marker occurs between the two elements that have a dependency relation.
- Importantly, the first element can be either the Head or the dependent.

(1) a. nasa mesa-^M**ng** ^Hlibro b. ^Hlibro^M-**ng** nasa mesa [Tagalog]
 on table-L book book-L on table

Both a and b: 'the book on the table'

(2) a. va'uan ***(a)** kun b. kun ***(a)** va'uan [Paiwan]
 new L skirt skirt L new

Both: 'new skirt'

(3) a. [k-in-asengseng ni Kai] ***(a)** kun
 make-PV GEN Kai L skirt
 b. kun ***(a)** [k-in-asengseng ni Kai]
 skirt L make-PV GEN Kai

Both: 'the skirt that is made by Kai'

- When the neutral marking is seen in nominals, the marker is called a **linker**. We call such a dependency marker in other categories also a **linker**.
- Moreover, a linker may have various forms depending on the formal context (Chung 1991), e.g., whether it follows a vowel.

a. i Chapanis **na** linguahi b. i linguahi-**n** Chapanis [Chamorro]
 the Japanese L language the language-L Japanese
 both a and b: 'the Japanese language'

- **A coordinator** occurs between two conjuncts, regardless of the structural hierarchy of the conjuncts (in both the left- and right-branching structures).
- **A MOD marker** also occurs between a modifier and the modified element (in both the left- and right modifier structures).
- They are dependency markers of the neutral pattern.

The element that realizes J may have clitic properties.

- Clitic: bound form in a phrase
proclitic $\approx$ prefix enclitic $\approx$ suffix
Its position is not purely syntactic.
- If an element is a clitic, we need to identify not only its position in the whole construction, but also its position with respect to its phonological host, i.e., whether it is a proclitic or enclitic.
- **Coordinators**: In English, a coordinator takes the conjunct to its right as its morphological host, whereas in Japanese, a coordinator takes the conjunct to its left as its morphological host.
- **MOD markers**: In Mandarin, the MOD marker *de* is an enclitic.

- (1) a. Baoyu shuo-**de** hen man. b. Baoyu hen man-**de** shuo.
Baoyu speak-MOD very slow Baoyu very slow-MOD speak

Both: 'Baoyu spoke slowly.'

Seventh, if there is only one coordinator or MOD marker and more than two items, it must be next to the final item.

Coordinator

(*and*: proclitic)

- (1) a. I saw Iris, Audrey, and Doris.
b. *I saw Iris, and Audrey, Doris.

MOD marker

(*de*: enclitic)

- (2) a. Alan mai-le yi-ge hen xiao, hen qing, hen bo-de shouji.
Alan buy-PRF one-CL very small very light very thin-MOD cellphone
'Alan bought a very small, very light, and very thin cellphone.'
b. *Alan mai-le yi-ge hen xiao, hen qing-de, hen bo shouji.

In addition to the seven shared properties of coordinators and MOD markers, not surprisingly, the two kinds of formatives may share the same form in some languages (see Zhang 2023: 35 for the references).

- (1) a. xia liang **er** dong wen. [Classic Chinese]
 summer cool and winter warm
 ‘Summer is cool and winter is warm.’
- b. Zilu shuai’er **er** dui.
 Zilu carelessly MOD answer
 ‘Zilu answered carelessly.’
- (2) a. S<um>’an eu’ bauwak ‘i’ Tapas [**ru**’ hab-un=nia’]. [Mayarinax]
 breed<AV> ACC wild.hog NOM Tapas LK kill-PV=3.SL.OBL
 ‘Tapas bred hogs and then killed them.’
- b. S<um>’an [**ru**’ pahab] cu’ bauwak ‘i’ Tapas.
 breed<AV> LK kill ACC wild.hog NOM Tapas
 ‘Tapas bred hogs to kill.’

4. The CSC Issue

- Ross (1967): Coordinate Structure Constraint (CSC):
In a coordinate structure, no conjunct may move, and nor may any element move out of a conjunct.
 - Both parts can be violated:
 - ▲ Conjuncts can move in languages such as Japanese and Serbo-Croatian (Oda 2021, Bošković 2022).
 - ▲ Elements can also move out of a conjunct:
- (1) a. What kind of cancer can you eat herbs and not get _?
 b. What kind of herbs can you eat _ and not get cancer?
- The CSC might be a processing constraint to a certain extent (Kehler 2002, Zhang 2010, Altshuler & Truswell 2022; also see Sportiche 2024).

5. Three Types of the ATB Constructions

- The Across-The-Board constructions:
 - (1) I wonder who John saw _ and Bill hit _.
 - (2) Which dog did Bill kill _ on Monday and Fred kill _ on Tuesday, respectively?
Possible answer: Bill killed Lucy and Fred killed Charlie.
 - (3) It was a total of \$3000 that I borrowed _ and my sister stole _ from the bank.
- Coordination is fascinating!
- Propose your own analysis and then compare it with Zhang's (2023) analysis of the three ATB constructions.

6. Significance

- Why J?

The function of **J**: it introduces a non-argument expression to a lexical element.

If an XP is the complement of J, it is a non-argument to the categorizer of the J-set.

There should be no labeling problem for a modification construction.

- There should be no construction-specific syntax.

No &P or ModP is necessary (cf. *toughP, *cleftP, *extrapositionP, *middleP...).

Neither complementation nor categorization is construction-specific.

- There is no unified syntax of coordination: left-branching and right-branching.

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